

Management of Huanglongbing of Citrus: Lessons from São Paulo and Florida

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Keywords

citrus, psyllid, *Candidatus Liberibacter asiaticus*, disease management, *Diaphorina citri*

Abstract

São Paulo, Brazil, and Florida, USA, were the two major orange production areas in the world until Huanglongbing (HLB) was discovered in São Paulo in 2004 and Florida in 2005. In the absence of resistant citrus varieties, HLB is the most destructive citrus disease known because of the lack of effective tools to reduce spread of the vector, *Diaphorina citri* (Asian citrus psyllid), and transmission of the associated pathogen, *Candidatus Liberibacter asiaticus*. In both countries, a three-pronged management approach was recommended and begun: planting only disease-free nursery trees, effective psyllid control, and removal of all symptomatic trees. In Brazil, these management procedures were continued and improved and resulted in relatively little overall loss of production. In contrast, in Florida the citrus industry has been devastated with annual production reduced by approximately 80%. This review compares and contrasts various cultural and pest management strategies that have been used to reduce infection by the pathogen and increase tolerance of HLB in the main orange-growing regions in the world.

INTRODUCTION

Two major orange juice production areas in the world have both been hit with outbreaks of the bacterial disease Huanglongbing (HLB) on citrus, but the course of the disease in these areas has diverged because of differences in disease management. HLB was first discovered in São Paulo, Brazil, in 2004 and in Florida, USA, in 2005 and has spread throughout both regions (**Figure 1**). In the São Paulo and Triângulo/Southwest Minas Gerais (SPMG) citrus belt, the region responsible for approximately 79% of Brazilian sweet orange production, the average incidence of HLB-symptomatic orange trees reached 38% in 2023 (36), but orange production in the SPMG was reduced by only about 10% during the five-year periods 2004–2008 (average production of 342.2 million boxes) and 2019–2023 (average production of 308.4 million boxes) (37, 38). In contrast, although citrus bacterial canker and urbanization had reduced Florida orange production by 38% between 1997 and 2005, HLB reduced production by 73% by 2021 and 90% by 2022 when compared to 2005 levels (110) (**Figure 2**). In both regions the yield loss was caused by an increase in premature fruit drop (**Figure 3**). Although Florida losses are partially due to hurricanes, this is a story of how HLB management procedures work in different environmental, cultural, and geographical settings—and how they do not.

To have HLB disease, three things must be present: a susceptible host, in this case citrus; a pathogen, the bacterium *Candidatus Liberibacter asiaticus* (CLas); and a vector, the Asian citrus psyllid (*Diaphorina citri*) (30). The psyllid is the host of CLas, and citrus is the vector that allows the bacterium to move to the next generation of psyllids (40).

The psyllid has been present in Brazil since the 1940s (64), but HLB was first reported in São Paulo State in 2004 (27, 104). Initially, a new related bacterium, '*Candidatus Liberibacter americanus*', was found in most citrus trees showing typical symptoms of HLB (105). However, within four years, CLas became the predominant species and was present in more than 99% of all analyzed HLB-positive samples (65).

In 2008, the first survey of HLB incidence in orange orchards in SPMG estimated an average of 0.61% symptomatic trees. From 2008 to 2015, the disease incidence nearly doubled each year until 2015, which represents an average annual increase rate of almost 50%. From 2015 to 2018, annual increases in HLB incidence stabilized between 16% and 18%. From 2018 to 2022, the disease



Figure 1

São Paulo and Triângulo/Southwest of Minas Gerais, Brazil, and Florida, USA. Images from Google Earth.

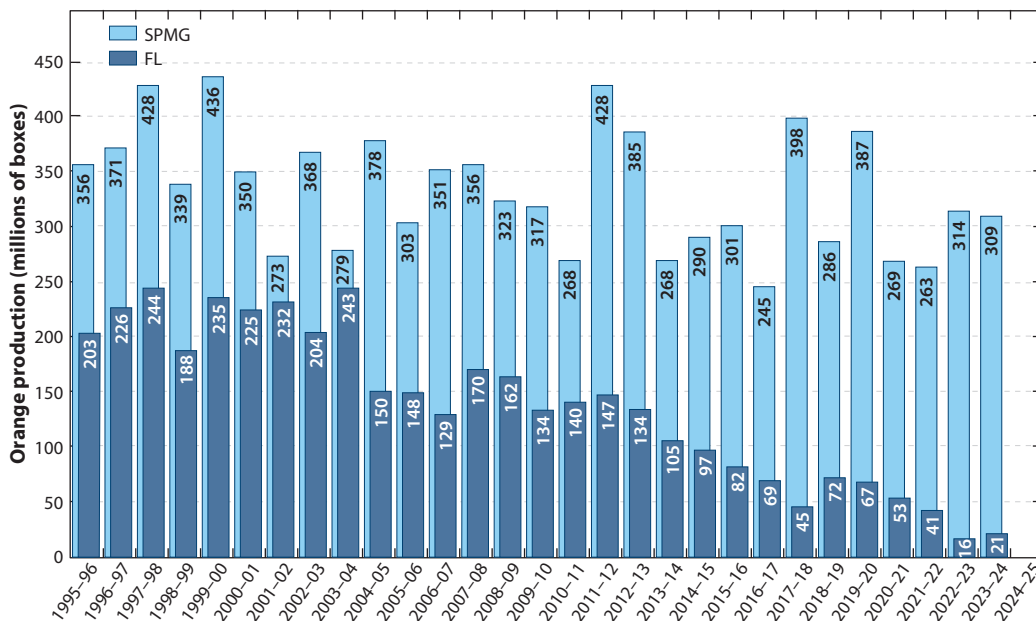


Figure 2

Orange production in São Paulo and Triângulo/Southwest of Minas Gerais (SPMG) and Florida (FL) from 1995–1996 to 2023–2024 (25, 35, 109).

incidence increased at an average annual increase rate of 8% (8). In 2023, a dramatic increase of 56% in HLB incidence was observed, reaching 38% disease incidence (36) (**Figures 3 and 4**). This increase was caused by the accumulation of diseased trees in commercial orchards and failures of psyllid control, mainly associated with psyllid resistance to the most used insecticides.

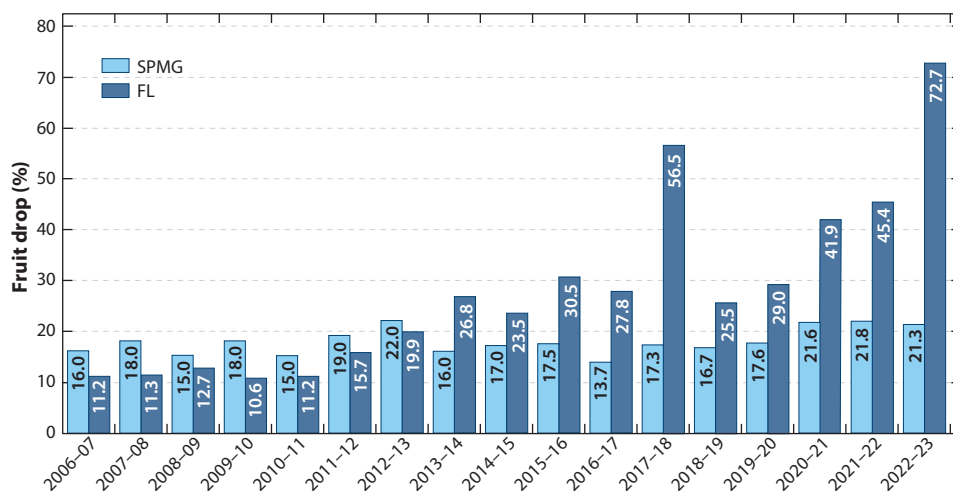


Figure 3

Fruit drop of orange in São Paulo and Triângulo/Southwest of Minas Gerais (SPMG) and Florida (FL) from 2006–2007 to 2022–2023 (25, 35, 109).

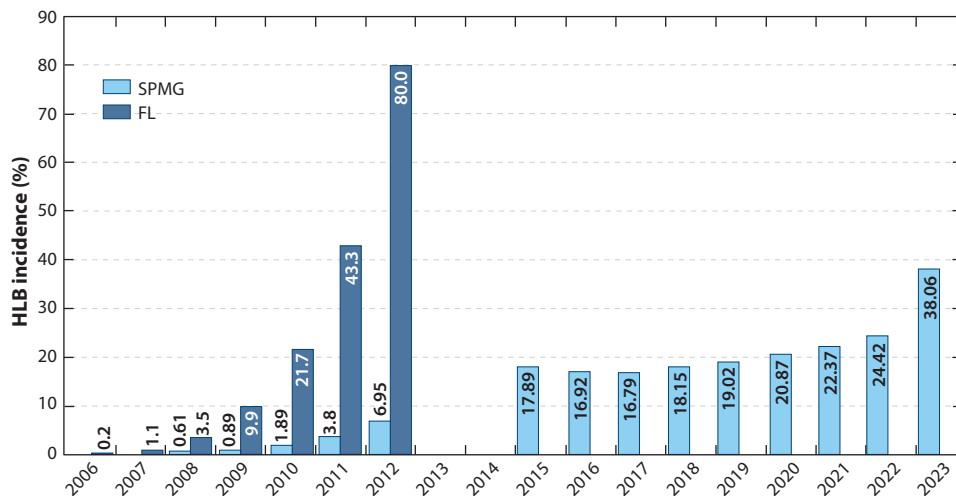


Figure 4

Incidence of Huanglongbing (HLB)-affected orange trees in São Paulo and Triângulo/Southwest of Minas Gerais (SPMG) and Florida (FL) from 2006 to 2023 (35).

In Florida, Dr. Susan Halbert, an entomologist working for the Florida Department of Agriculture and Consumer Services, Division of Plant Industry (FDACS-DPI), discovered the psyllid in Palm Beach County in June 1998 (49). Within two years, the psyllid was transported throughout Florida on citrus and orange jasmine (*Murraya paniculata*) host trees propagated in South Florida nurseries and sold at garden centers (41, 49). In 1999, a detection of CLas in the psyllid was reported but not confirmed by independent testing (49). At that time, there should have been a recognition that infected citrus plants were present in South Florida based on the high probability that the pathogen was introduced with illicit plant material from Asia (40), but there was not.

Dr. Halbert's survey eventually led to the discovery of HLB trees in a fruit and vegetable farm in Homestead in southern Miami-Dade County in August 2005. In the months following, delimiting surveys by FDACS-DPI detected HLB 160 miles north in residential St. Lucie County and 100 miles northwest in the major citrus production area of Hendry County. Hence, HLB was present in South Florida for as much as 4–6 years prior to its discovery. Given the extent of HLB in Florida, the latency of the disease symptoms, and citrus grower resistance, FDACS-DPI considered eradication of the disease impossible (49).

In the first years after 2005, disease progress was estimated by FDACS-DPI surveys to be spreading north at 36 miles per year (66). Based on data from Southern Gardens Citrus Diagnostic Lab (SGCDL) in Clewiston, Florida, HLB incidence was 0.2% in 2006, 1.1% in 2007, and 3.5% in 2008 and doubled every year after to 9.9% in 2009, 21.7% in 2010, and 43.3% in 2011 (53) (Figure 4). Assuming another doubling in 2012, HLB incidence was ~80% by 2013 coincident with a rise in HLB-induced fruit drop (Figure 4, Figure 5a). The steep decline in tree productivity since 2013 supports this estimation (94) (Figure 2).

WHAT WAS KNOWN ABOUT HUANGLONGBING AT THE TIME OF THE INVASIONS INTO SÃO PAULO AND FLORIDA

As HLB entered São Paulo and Florida, there was a limited amount of information available concerning this disease. Citrus diseases like HLB had been known to occur in Asia and South Africa

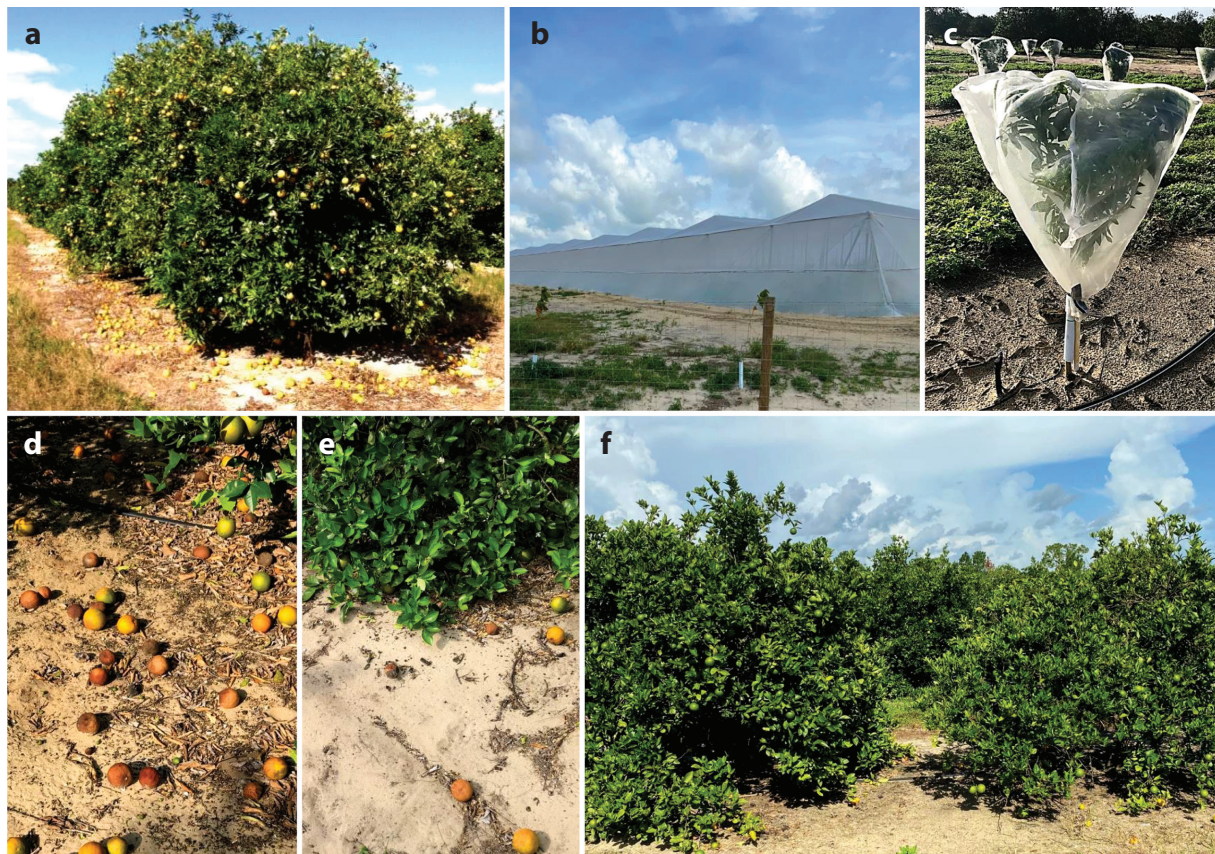


Figure 5

(a) Fruit drop from 20-year-old Valencia orange trees at Lake Alfred, FL, in 2013. (b) Citrus under protective screen for psyllid exclusion. (c) Individual protective covers for psyllid exclusion. (d) Fruit drop from untreated 12-year-old Valencia orange trees in February 2022. (e) Fruit drop from untreated 12-year-old Valencia orange trees in February 2022 that received foliar applications of GA3 (ProGibb LV) at 50 g active ingredient (a.i.)/hectare in September, October, and November 2021. (f) Hamlin orange trees treated in July 2022 by trunk injection of 0.7 g of oxytetracycline (OTC) (39.6% a.i.) on each side of the trunk (40 mL total per tree) showing signs of greener and thicker canopy (left) compared to untreated trees (right). Harvested fruit have more than a one-unit increase in pounds of solids (4.3 versus 2.99 OTC in treated versus untreated).

for a long time and had been associated with psyllids. The Asian disease was thought to be transmitted by *Diaphorina citri*, and the South African disease was transmitted by *Trioza erytreae*. The South African psyllid and disease were found to be heat sensitive, whereas the Asian psyllid and disease were not heat sensitive (17). In 1970, J.M. Bové & Dominique Laflèche (61) found that the diseases were associated with bacteria. Monique Garnier in Bové's laboratory (39) then demonstrated that the bacteria had cell walls and were gram-negative. In the Bové lab in 1996, the 16S ribosomal DNA fragment was cloned and developed into a PCR assay for the associated bacterium (55).

J.M. (Josy) Bové, who was the most experienced and knowledgeable scientist concerning HLB, immediately traveled to São Paulo as soon as the pathogen was discovered to advise how to manage this disease (18). Similarly, he also led Florida scientists in reducing the spread there.

MANAGEMENT STRATEGY PROGRESSION IN BRAZIL

In São Paulo, initial management strategies began with the three-pronged approach recommended by Bové (18): planting only disease-free nursery trees, effective psyllid control, and removal of all symptomatic trees. Until 2009, control measures focused almost exclusively on commercial orchards. The initial accomplishment of the three-pronged system to prevent the infection of new trees (17) by most citrus growers of Brazil was facilitated by their previous experience in combating citrus canker and citrus variegated chlorosis (CVC), which also required the planting of healthy nursery plants produced in protected nurseries since 2003, frequent inspection and elimination of diseased trees, and preventive insect vector control (CVC sharpshooters).

Starting in 2005, federal and state regulations for mandatory removal of HLB-symptomatic trees were established and undertaken by the Plant Protection Agency Coordination of São Paulo State (Coordenadoria de Defesa Agropecuária do Estado de São Paulo) with the support and partnership of the growers and juice manufacturers association Fundecitrus (84). At least four inspections per year to detect HLB-symptomatic trees and immediate removal of symptomatic trees by growers in commercial orchards were required, but the law was not enforced by the State Plant Protection Agency Coordination. Growers' voluntary adherence to the removal of symptomatic trees began to diminish gradually after 2010, and the HLB Eradication Program was not sufficient to prevent HLB from spreading. It became evident that the elimination of HLB-symptomatic trees from commercial orchards, when adopted alone by a given grower located in a region with a high incidence of the disease and/or whose orchard was relatively small, had little effect on reducing disease progression within an orchard. This happens because removal of inoculum sources within an orchard plays an important role in curbing the spread of secondary infections but has no effect in suppressing primary infections caused by psyllids that acquired the pathogen in inoculum sources outside the commercial orchards (9, 10).

Approximately 64.1 million orange trees with symptoms of the disease were removed between January 2005 and June 2022 (28). Until 2013, the number of eradicated trees with HLB grew year after year and was similar to the number of HLB-affected trees. Then, after 2013, the difference between eradicated trees and HLB-affected trees gradually increased each year. This resulted in the gradual accumulation of HLB-symptomatic trees, especially in orchards with adult fruit-bearing trees that have a high HLB incidence. This situation promoted a change in the HLB federal and state regulations in 2021: the mandatory elimination of HLB-symptomatic trees up to 8 years old in commercial orchards. After this age, elimination becomes optional as long as psyllid control is carried out to avoid the proliferation of psyllids and bacteria acquisition in diseased trees. Also, the mandatory removal of symptomatic trees was expanded to noncommercial orchards (Ministry of Agriculture, Livestock and Supply, Ordinance 317 of 05/21/2021 and Secretary of Agriculture and Supply, Resolution 88 of 12/07/2021). Although not enforced with the necessary rigor, the Eradication Program was very important in slowing down disease progression in the first decade of the HLB epidemics and has continued to be important in maintaining low disease incidence in other Brazilian citrus areas having low disease incidence.

With the accumulation of symptomatic trees in the orchards, the control of HLB became totally dependent on the efficacy of psyllid control, and the frequency of systemic and contact insecticide sprays has increased from the onset of HLB in SPMG to the present day because of the difficulty in preventing infections during new shoot development. The systemic insecticides are applied to the trunk surface or as a soil drench in trees younger than two years old, and contact insecticides are applied as foliar sprays (23, 31, 70, 71, 111). Currently, drench application of systemic insecticides is recommended one to five days before planting nursery trees, and three to four applications per year to the leaf flushes during the rainy season from late July to the end of March

for trees up to three years old are recommended. Foliar applications at intervals of 7–10 days are recommended for both young and mature trees. However, even with frequent application of insecticides, it is almost impossible to completely avoid new infections when bacterialiferous psyllid populations are constantly arriving in the orchard.

Growers in Brazil that sell fruit to the juice industry can only use insecticides from the ProteCitrus list (<https://www.fundecitrus.com.br/protectcitrus>) to comply with regulations of orange juice–importing countries. Due to restrictions on use, especially from the European Union, additional insecticides for controlling the psyllid continue to be banned every year, reducing the options for rotation of insecticides that use different modes of action. Therefore, in the past two years, the increase of resistance in psyllids to the least expensive and most used insecticides has compromised the efficacy of vector control, is associated with a high number of diseased trees in orchards, and was responsible for the significant increase in HLB incidence in 2023.

To complement psyllid control and as part of antiresistance strategies, processed kaolin, which repels the psyllid and interferes with its feeding (69, 72), and entomopathogenic fungi targeting the psyllids (7) have been developed and used in the field. Another control measure that resulted in the reduction of psyllid infestation and dispersal is based on the push–pull and kill strategy. In this case, processed kaolin is sprayed on the citrus crop as a psyllid repellent (push), and psyllid-attractant curry (*Bergera koenigii*) trees (pull) treated with insecticides (kill) are planted at the citrus orchard border as a lethal trap crop (32).

Protecting citrus orchards with insecticides was found to be ineffective if infected psyllids could fly in from infected trees (16, 47). Therefore, with the development of the Phytosanitary Alert program (<http://www.fundecitrus.com.br/alerta-fitossanitario>), Fundecitrus organized groups of citrus growers for regional psyllid monitoring and to coordinate simultaneous insect control in all the commercial citrus orchards in a region. In this program, growers voluntarily monitor the psyllid population every two weeks with yellow sticky cards in their commercial orchards, and the Fundecitrus team monitors small orchards and sites with poor or no psyllid management. Based on the growing trend of psyllid population and the predominant citrus vegetative stage, alerts are sent to all citrus growers in the region, whether or not they are participating in the data collection, because due to constant psyllid dispersal among citrus orchards, better control is obtained where all growers apply insecticide at the same time. The alert guides citrus growers to carry out a coordinated area-wide spray in commercial citrus orchards, usually within a week.

Because citrus trees present in abandoned orchards, noncommercial orchards, backyards, pastures, and woodlands that could be infected by HLB do not receive the area-wide insecticide applications recommended by the Phytosanitary Alert, actions beyond those on commercial citrus farms are necessary to complement the control of these primary inoculum sources (15). Protected Areas with one or more neighboring citrus orchards have been voluntarily established throughout the SPMG and became part of Fundecitrus's HLB Combat Program (56, 68, 80). In this program, the Fundecitrus team locates and negotiates with property owners to eliminate the citrus and orange jasmine trees present in a radius of 5 km around the commercial citrus properties located in the Protected Area. The partner citrus growers are responsible for the elimination of HLB host plants and their replacement. In the case of citrus trees in pasture or woodland, the inspection to locate the citrus trees and the removal of them are the responsibility of citrus growers, not Fundecitrus. In the case of abandoned orchards, negotiations are made to eliminate the entire orchard, with a strong recommendation to spray insecticide before tree removal to prevent the spread of bacterialiferous psyllids present on tree canopy to neighboring commercial orchards. If tree removal is not possible, neighboring growers come together to support or even apply insecticides or release *Tamarixia radiata* in poorly HLB-managed orchards. From 2017 to 2022, more than 5.1 million citrus trees were removed from abandoned orchards, pastures, and backyards with an

effectiveness of 97%; that is, for each 100 citrus trees found in abandoned orchards, pastures, and backyards, the Fundecitrus team was able to remove 97 trees.

MANAGEMENT STRATEGY PROGRESSION IN FLORIDA

In Florida, the three-pronged approach employed in Brazil (14) was recommended by the United States Department of Agriculture, Agricultural Research Service (USDA-ARS) and University of Florida plant pathologists. To meet the requirement for disease-free nursery trees, discussions among citrus nurserymen, production managers, and University of Florida scientists led to legislation of a revised Florida statute in 2006 for enclosing Florida nursery structures to protect nursery trees from vector transmission (95). The new rule required the propagation of nursery trees and budwood source trees in insect-proof screenhouses along with other measures to keep budwood and nursery stock from spreading CLAs. By January 2008, production of all nursery trees in enclosed greenhouses was achieved but required significant capital expenditures by nursery operations and a doubling in price for nursery trees. The lack of documented cases of HLB introduction into new plantings via infected stock is evidence that this leg of the three-pronged approach was successful (106).

In the wake of the citrus canker eradication program from 1985 to 2005 that resulted in the destruction of millions of nursery and producing trees, many Florida growers gave only passing consideration to the tree removal component of the three-pronged approach. However, between 2005 and 2011, a few growers stringently followed the recommendations (42). Accurately identifying incipient leaf mottling symptoms typical of HLB before tree removal required lengthy training of ground survey crews. Until inspectors became experienced, SGCDL performed quantitative PCR assays to confirm leaf symptom detections for the presence of HLB at no cost to the grower (54). However, stringent removal of infected trees occurred for only a few years because spread of CLAs was so rampant that continued removal would have resulted in no trees left.

Instead of tree removal, citrus operations statewide relied on vector control. Although growers readily adopted psyllid control measures, CLAs had spread to 100% of individual orchards in Florida by 2013. Incidence of bacterialiferous psyllids rose steadily, and the rate of vector transmission accelerated disease spread more rapidly than expected because of underestimation of the distance that psyllids disperse (16, 67) and the high efficiency of CLAs acquisition by feeding nymphs on emerging leaf flush (62). Without the elimination of primary CLAs inoculum in the form of infected trees (10), psyllid control was wholly inadequate for the reduction of disease spread.

Citrus growers applied as many as 12 insecticide sprays for psyllids (often more for young plantings) on a calendar-based cycle in addition to dormant-season spraying. However, this intensive usage of insecticides was not sustainable economically nor compatible with biological control by psyllid predators and parasitoids. Inevitably, entomologists detected psyllid insecticide resistance by 2009 (107).

Alternative methods for psyllid management included biological control (82), cultural control with insect-repellant reflective mulches (29), and mitigation of psyllid ingress by natural and artificial windbreaks (67). None of these tactics proved as effective as dormant season applications of insecticides (83).

Before HLB became endemic in Florida, a threshold within the framework of IPM was not applied (81). More recently, a spray program of rotated insecticides based on an economic threshold of 0.5–1.0 adult psyllids per foliage sample was found to potentially provide economic benefits as well as benefits with respect to conservation of pest predators and parasitoids to prevent psyllid damage to foliage that causes debilitation of tree health (24).

By 2010, based on the recognition that psyllid populations were not effectively controlled by individual growers due to regional dispersal of the vector, Citrus Health Management Areas (CHMAs) were set up (92). CHMAs coordinated psyllid sprays to reduce movement of psyllids between orchards and shorten the time required for follow-up sprays. The CHMA program was supported by a multiple pest survey developed by the USDA-ARS in collaboration with the USDA, Animal and Plant Health Inspection Service, Plant Protection and Quarantine (79). The survey used a risk-based algorithm to select sites for assessment of commercial citrus blocks to estimate both psyllid population distribution and CLAs-infected tree incidence. The multi-pest survey and scouting reports of psyllid populations on the CHMA website served as a tool for making pest management decisions. However, by 2017, interest in CHMAs began to wane because of the continued decline in yields, even in the best CHMAs, even though they maintained remarkably low psyllid populations for years. Owing to declining participation by citrus growers, CHMAs are no longer active in Florida (<https://crec.ifas.ufl.edu/extension/chmas/>).

CITRUS UNDER PROTECTIVE SCREEN PRODUCTION IN FLORIDA

Psyllid exclusion remains the most effective strategy to keep citrus trees free of HLB. In Florida, citrus under protective screens (CUPS) has been proven effective for growing full-size trees in 4 ha screen-enclosed structures to keep them disease-free (91) (**Figure 5b**). The benefits of eliminating HLB in such an exclusionary planting include rapid, normal tree growth, high production of quality fruit, negligible fruit drop, and calibrated fertilizer and irrigation requirements. However, this system is expensive and probably most applicable for specialty crops, such as fresh-grapefruit varieties that have a high return on investment (34, 85). The CUPS significantly increases the cost of establishing a citrus orchard due to the high cost of screen-house construction (an average of \$11 per square meter). The 40- to 50-mesh screen may need replacement every 7 to 10 years, depending on exposure to tropical storm winds, although replacements come at a lower cost than the entire initial structure. Nevertheless, a variety of pests are significantly reduced, and their infestations are localized and easy to manage when detected early. However, the environment in the CUPS promotes development of citrus rust mite (*Phyllocoptura oleivora*) and Chilli thrips (*Scirtothrips dorsalis*). Greasy spot, caused by *Zasmidium citri-griseum*, and other fungal diseases may also thrive in the more humid conditions of the screen-house environment. Although CUPS results in high-quality fruit production, its high cost has limited this production to about 500 acres in Florida.

INDIVIDUAL PROTECTIVE COVERS IN FLORIDA

In mature trees already infected with CLAs, the main goal is to sustain fruit production. In contrast, young trees are HLB free at planting, and the goal is to keep them disease free as long as possible, usually at least until they bear fruit. Individual protective covers (IPCs) are a novel strategy introduced in Florida in which psyllids are excluded from individual trees using a protective mesh bag that covers the tree for the first 3–4 years in the orchard (2) (**Figure 5c**). IPCs are economically more feasible than CUPS structures for processing varieties grown on a large acreage in Florida and have been widely adopted by growers to delay infection of young trees for 3 to 4 years.

LIVING WITH HUANGLONGBING

As it became evident that prevention of the spread of HLB in Florida was failing, management strategies focused more on efforts to reduce levels of disease symptoms in infected trees. Early examples of such strategies were attributed to improved nutritional programs, which led to growers all over the state experimenting with fertilization. Initially, the nutritional programs were based

on foliar sprays of micronutrients (86, 89) and host disease resistance inducers (46). Although nutritional programs improved tree appearance (96), documentation of increases in the yield of HLB-affected trees was lacking (57, 97). Trees continued to decline until programs employed fertigation (fertilizer injection into an irrigation system) with readily available forms of macro- and micronutrients on weekly or biweekly schedules (75–77). These nutritional regimes along with frequent irrigation cycles of shorter duration were formulated to mitigate stress on declining trees by compensating for the reduced density of the fibrous root system on HLB-affected trees (58). These strategies worked because HLB-damaged roots take up readily available nutrients from the soil solution as effectively as healthy roots (89).

However, one result of the more intensive pesticide and fertilization programs was that the cost of production rose substantially. Costs for growing and processing oranges increased from a range of \$757–\$985 per acre in 2004–2005 to at least \$1,681 in the 2014–2015 season. The cost of the foliar insecticide spray program alone increased by almost 260%, and the cost of fertilizers and their application increased by 134%. Similarly, the 2003–2004 production cost for fresh market grapefruit on the Florida east coast (Indian River region) was \$1,367 per acre. By 2017, the increase in the cost of foliar fungicide and insecticide sprays and fertilizer programs for grapefruit was 164% and 137%, respectively (90). Although fruit prices have risen, this increase has not been enough in most cases to compensate for the increased production costs (91).

In São Paulo, enhanced nutritional programs with macro- and micronutrients, phytohormones (auxin), and potential resistance inducers such as phosphites and salicylates did not reduce disease severity or maintain high yields in affected trees (11–13). These contrasting results in comparison to Florida could be partially explained by the higher soil fertility and good standard nutrition programs in São Paulo, with current micronutrients commonly already applied and soil acidity tightly controlled by applications of lime.

Field transmission of the pathogen to trees by psyllids results in a prolonged incubation period of 6–12 months when the disease remains subclinical prior to visual symptom expression. After infection, decline of the root system occurs while the canopy remains asymptomatic (43, 44, 58). Asymptomatic trees with detectable CLAs in fibrous roots (less than 2 mm in diameter) already sustain substantial root loss compared to healthy trees. Loss of fibrous root mass due to HLB was equal to or greater than 27% in the surveys of young trees (3–4 years old) and older trees (10–25 years old) alike (43, 44).

This loss of fibrous roots is associated with a shortened lifespan of the roots. Infected roots have a lifespan of only 1.5–4 months, whereas healthy roots live for 9–12 months. The shortened lifespan is accompanied by increased root growth, leading to an imbalance in carbohydrate demand in the tree while still reducing the total uptake capacity of the root system. Structural roots (greater than 2 mm in diameter) also die on HLB-affected trees. Currently, there is no proven management option for prevention of HLB-associated root loss. Treatments that stimulate root growth are not recommended because they may further increase the root–canopy imbalance, resulting in further decreases in fruit yield. Instead, the recommendation has been to increase the frequency of soil nutrient and water applications to compensate for the limited uptake capacity of roots and to maintain or improve root longevity. This includes adjustment of soil pH to between 5.8 and 6.5 to avoid additional stress on the root system (76).

Owing to the massive loss of fibrous roots, maintaining the health of the root systems of HLB trees in orchards usually requires site-specific management. The initial step is to identify the stresses present in the orchard. In Florida, bicarbonates are leached from limestone in aquifers and act as a buffer, raising the pH of the water and irrigated soils. Soil pH and bicarbonates in irrigation water have gained attention in HLB management because HLB has reduced the tolerance of rootstocks such as Swingle citrumelo (*Citrus paradisi* × *Poncirus trifoliata*) planted on high soil

pH and/or irrigated with water high in bicarbonates and salts (74). Management of soil pH and bicarbonates with soil-applied sulfur and treatment of irrigation water with injections of sulfuric acid resulted in increased root density of trees with HLB. For Swingle citrumelo rootstocks, the ideal pH range is 5.5–6.5. Recent field experiments have determined that maintaining soil pH in this range increases nutrient uptake and root density (74). pH should be tested before and after treatment because overacidification risks the release of copper and other metals at toxic levels and can result in depletion of essential nutrients such as calcium and magnesium from the soil. The use of elemental sulfur for acidification is dependent on microbial oxidation, which can take a year or more to reduce soil pH. Soil tests for nutrients such as calcium and magnesium are used to guide supplemental application of nutrients lost from leaching and prevent copper toxicity to roots (45). When pH management is necessary, gypsum (calcium sulfate) is used to supply calcium without increasing soil pH.

PLANT GROWTH REGULATORS FOR MANAGEMENT OF PREMATURE FRUIT DROP IN FLORIDA

Preharvest fruit drop is a symptom and major cause of economic loss induced by HLB in Florida. Because fruit cannot be harvested from the ground, fruit drop can represent a loss of a large portion of the yield. In 2012–2013, fruit drop losses in Florida were 18% for early and mid-season varieties and 22% for late-season varieties. In 2013–2014, fruit drop losses exceeded 30% in some citrus cultivars. This unprecedented fruit drop is attributed to root loss associated with HLB (44, 95). Currently, fruit drop in Hamlin and Valencia sweet oranges can be as high as 50% of the crop (**Figure 3**). In contrast, Sugar Belle mandarin experiences minimal mature fruit drop and no reduction in yield (98). In this cultivar, several metabolites conferring protection against oxidative stress were found in high amounts (59), suggesting a link between oxidative stress and low fruit drop (102). Interestingly, higher zinc levels in trees are associated with lower fruit drop associated with HLB. Zinc is a micronutrient that is involved in both modulation of oxidative stress and auxin homeostasis in plants (21, 100). The timing of zinc application affects fruit drop, which suggests that zinc modulates auxin and/or oxidative stress detoxification. Hence, zinc treatments result in higher final yields (78). Potassium acts to boost the effect of zinc in reducing fruit drop. Potassium increases fruit size and retention, supporting previous findings showing an inverse relationship between fruit size and drop susceptibility (102, 103).

Gibberellic acid (GA3), when applied as early as September through January, increases fruit detachment force (likelihood of fruit to be retained on tree) along with yield and fruit size (**Figure 5d,e**). This contrasts with previous findings (1, 48). This disparity may be due to application timing. In a recent study that demonstrated reduced fruit drop, GA3 was applied approximately 8 months before anticipated harvest, when the fruits were still in the development phase.

Auxin levels also modulate fruit drop. 2,4-Dichlorophenoxyacetic acid (2,4-D) is a synthetic auxin often applied to trees to alter auxin-mediated phenotypes. The two main uses of 2,4-D application in citrus production are preharvest fruit drop control and to increase fruit size. The reduction in preharvest fruit drop in citrus by applying foliar sprays containing 2,4-D has been well documented since 1947 (26). 2,4-D has been shown to increase fruit size when applied 6–8 weeks after flowering (33). However, there are limited studies on the efficacy of 2,4-D in reducing HLB-associated preharvest fruit drop. Albrigo & Stover (1) reported an inconsistent efficacy of 2,4-D in controlling fruit drop in HLB-affected trees. However, these applications were performed close to the time of harvest. Based on the existing literature on fruit drop control, both 2,4-D and GA3 can potentially control HLB-associated fruit drop when applied at an early stage of fruit development (102).

The application of brassinosteroids (Brs) to protect young citrus trees against HLB has been reported (22). Although Br treatment increases resistance to a wide range of pathogens in plants, the role of Brs in plant responses to pathogens is complex and the regulation of immunity occurs at multiple levels (60). In citrus affected by HLB, it has been reported that some genes associated with systemic acquired resistance, such as chitinase and β -1,3-glucanase, are enhanced by Br treatment (3, 22). Recently, it was discovered that Brs applied to newly planted trees significantly delay HLB disease development, protecting trees for several months after biweekly application as compared to nontreated trees. Furthermore, applications of Brs reduce psyllid colonization in new leaf growth and the incidence of other insect pests such as Citrus rust mites in fruit-bearing trees. These effects are likely due to the induction of a strong immune response in the citrus trees (22). However, the immune response triggered by Brs in citrus plants has not been fully characterized, and it is not known how long the priming effect lasts after treatment (5).

In mature fruit-bearing trees (6–7-year-old trees) that are already HLB positive and declining in production, an improvement in yield and internal fruit quality occurs after foliar Br spray treatment. Biweekly Br spray also reduces fruit drop, resulting in a 10–21% increase in yield (2). Thus far, just one application before the onset of maturation of Valencia fruit induces a significant increase in sugar and the ratio of sugar to acid. Determining the duration of disease resistance of citrus trees to Brs dictates the optimum number of treatments required to achieve the best protection. In addition, determining the best timing during maturation for application of Br to induce sugar accumulation in fruit should result in efficient treatments to increase sugar when needed.

ANTIBIOTIC THERAPY FOR MANAGEMENT OF HUANGLONGBING

Antibiotic experiments using tetracycline hydrochloride, an inhibitor of bacterial protein synthesis, against HLB by trunk injection were conducted in South Africa in the 1970s. Tetracycline treatments were also conducted many years ago in Reunion (6), Taiwan (99), and Indonesia (101). Even though remission of symptoms was observed, tetracycline injections were not a practical treatment because of their expense and only temporary symptom remission, as tetracycline is bacteriostatic, not bactericidal. Treated trees also exhibited phytotoxicity symptoms in the vasculature at the injection site, and residues were detected in the fruit (20, 73, 87, 88). Penicillin also had a strongly positive effect on HLB-affected greenhouse trees (19) as well as on field trees in Reunion Island (6). The efficacy of penicillin provided evidence for the presence of peptidoglycan in the cell wall of the HLB bacterium (19). However, because of the medical use of penicillin, it is unlikely to be approved for use on any agricultural crop.

In Florida in 2016, three products with either oxytetracycline (OTC) or streptomycin as an active ingredient were approved for emergency use by the EPA as a foliar spray. Recent studies have cast doubts on the efficacy of foliar application of antibiotics because they must enter the phloem to kill CLAs residing there (63, 108). Results from foliar applications have been inconsistent and have not proven to be an economically advantageous option for growers (93).

In 2022, trunk injection of one OTC product was approved for use in Florida. In a recent study of the uptake and distribution in citrus trees after pressurized injection, OTC was detected in the xylem, phloem, leaves, and roots within 48 h after trunk injection (52). To understand the mechanism of OTC uptake and translocation, the movement of OTC in girdled and nongirdled citrus seedlings and trees was followed after trunk injection (50). OTC was present in the phloem and xylem below and above the girdle, indicating that the xylem is the main route for OTC movement in the tree.

Trunk injection of OTC has been shown to reduce CLAs levels (51, 52) and improve plant health, fruit quality, and yield of young citrus trees (4) (Figure 5f). However, more research is

needed to optimally apply OTC, including the rate and timing of OTC application during the season.

WHAT MANAGEMENT STRATEGIES WORKED WHERE AND WHY

HLB entered São Paulo and Florida around the same time and initial management strategies were similar, but production in Florida has been reduced by approximately 80%, whereas in the SPMG production has been reduced in general by only about 10%. The three-pronged approach, namely control of psyllids, removal of sources of the bacterium, and prevention of transplanting infected trees, was employed in both countries, but it is instructive to address what led to differences in effectiveness in the two locations.

The third leg of the strategy appeared to work well in both countries. Procedures were established and enforced to produce nursery trees under protective covers in both countries, and there are no reports of planting HLB-infected trees.

Removal of infected trees did not work effectively in Florida. This was partially due to the memories of the recent canker eradication program, causing reluctance on the part of many growers to strictly adhere to a tree removal program. Although Florida growers sprayed their blocks, they ignored, and had little control over, external inoculum sources. Inevitably, even for growers who strenuously scouted and removed trees as soon as possible, the disease spread so rapidly that continued removal would have resulted in excessive removal of trees. In contrast, symptomatic tree removal has continued in SPMG while maintaining sustained production. This was partially due to the growers' experience with managing CVC and canker and enforcement by the Plant Protection Agency. However, the fact that it was working was the main reason it continued.

Notwithstanding SPMG's success, removal of symptomatic trees is not as effective in removing the source of inoculum for psyllids as expected. First, it is hard to identify all infected trees, which can function as a source of inoculum for noninfected psyllids that enter an orchard. Also, removal of infected trees does not prevent spread if infected psyllids enter an orchard from outside. When incoming psyllids settle on a noninfected tree with new flush, the female inoculates the leaf with CLAs while laying eggs. Many of the progeny nymphs acquire CLAs from the newly infected area of the leaf where they were born, and the resulting newly infected adults can immediately transfer the bacterium to additional trees.

Psyllid control was also very different in Florida and Brazil. In São Paulo, growers previously were experienced with the control of sharpshooters that transmit *Xylella fastidiosa*. They were keenly aware of the need to scout for the vector and apply insecticides in a timely way with the requisite capacity of manpower and sprayers. Early on, Fundecitrus researchers discovered and advised growers that control of psyllids outside of an orchard was at least as important as control within the orchard. This recognition led to the development of area-wide insect management strategies in both places. However, Brazilian growers adopted three-pronged management in a more organized and coordinated way than Florida growers. In Florida, some area-wide CHMAs were more effective than others, but eventually all were abandoned. In great contrast, area-wide management in Brazil evolved into the program of external management of not just psyllids but the host sources of the psyllid.

Geography also played an important role in the management of HLB. In Brazil, there are large areas without citrus where growers could move their orchards to isolate them from other HLB-infected orchards. Also, many farms consist of large contiguous areas where the grower has control of the whole area. In contrast, in Florida, most orchards are much smaller and border a neighbor's orchard. Not all neighbors are good neighbors with respect to pest management. Some growers in Florida have as many as ten different neighbor orchards.

Another factor is that Brazilian growers have alternative crops that they can grow economically. Small growers who can no longer produce citrus economically due to excessive HLB infection can switch to sugar cane, for example. Florida growers do not have an alternative crop that is likely to be economical to grow. For these growers, the alternative is to continue to grow HLB-infected citrus until it is no longer possible. For growers lucky enough to be located near suburban areas, they can sell land to developers to produce housing tracts. That is one reason for the large decrease in citrus acreage in Florida since the invasion of HLB.

A major reason that SPMG growers have sustained production despite the widescale removal of orchards was the adoption of the latest horticultural technology for matching rootstocks/scion combinations with appropriate tree spacing and by optimizing the use of irrigation and fertilization. The irrigated area in SPMG increased from 5% in 2000 to 36% in 2023, and the planting density increased from 388 trees per hectare in 2000 to 567 trees per hectare in 2022 (37). The result has been the production of higher yields on less land through more efficient and appropriate inputs of irrigation, fertilizer, and pesticides than in the past. In the best farms, production has doubled compared to production prior to the arrival of HLB. In Florida, most growers already had good irrigation and nutrition programs, so there was less opportunity to increase production.

THE LOOK FORWARD

The therapies that are being used in Florida—spray application of 2,4-D, GA3, and Brs and injection of oxytetracycline—appear to be reducing losses sufficiently to allow most remaining growers to continue farming for a few more years. However, because of the increased cost of production and the reduction in the size of the industry and associated infrastructure such as juice factories, this time is limited. Another potential band-aid solution that is moving toward regulatory approval is the use of the Citrus tristeza virus–based vector to deliver possible biotechnological therapies. This vector has been developed to express spinach defensin peptides that are continuously produced in the tree as the virus spreads and multiplies to directly inhibit CLAs in the phloem. An advantage of such a management strategy is that it is likely economical because the virus could be introduced into the young tree in the nursery with no further labor needed. However, this is not a permanent solution, as the inserted sequences can be lost from the virus within 7–10 years, resulting in only the wild-type virus.

Most growers have concluded that the only thing that can save the Florida industry in the long run is the engineering of a resistant or sufficiently tolerant tree. Genetically engineered trees—either as a genetically modified organism (GMO) or preferably as a non-GMO—are currently being developed. Transgenic scions that are tolerant of HLB are being tested in the field and regulatory approval is underway. As an example of the success of such strategies, genes have been silenced using CRISPR resulting in non-GMO canker-resistant citrus trees. The same methods are being used to produce non-GMO HLB-resistant or -tolerant trees. Such breakthroughs should result in sufficiently tolerant or resistant trees being available to growers within the next few years. However, the long transition from juvenility to maturity, the high cost of field testing, and nursery tree propagation are the main processes slowing this development. Thus, these trees will be useful only if the management strategies provide enough productivity to allow the Florida industry to survive until they are available; however, they could still be useful in other citrus-growing regions of the United States and the world that are affected by HLB. A major benefit of tolerant or resistant trees would be the reduction in costs of producing citrus in the presence of HLB. Even though the Brazilian citrus industry has not been reduced as much by HLB as the industry in Florida, the cost of production has substantially increased, making tolerant and resistant trees a major benefit to the industry there also.

To date, HLB has been a growing threat to Brazilian growers but has not become as much of an existential threat as in Florida. Helpful therapies have been developed, but a durable and sufficient solution for long-term survival is not yet available. Nevertheless, no stone is being left unturned as scientists and growers work hard to save these iconic industries.

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